

National Coverage Determination (NCD)

External Counterpulsation (ECP) Therapy for Severe Angina

20.20

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Tracking Information

Publication Number

100-3

Manual Section Number

20.20

Manual Section Title

External Counterpulsation (ECP) Therapy for Severe Angina

Version Number**Effective Date of this Version**

03/20/2006

Implementation Date

04/03/2006

Description Information

Benefit Category

Physicians' Services

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description**A. General**

External counterpulsation (ECP), commonly referred to as enhanced external counterpulsation, is a noninvasive outpatient treatment for coronary artery disease

refractory to medical and/or surgical therapy. Although ECP devices are cleared by the Food and Drug Administration (FDA) for use in treating a variety of cardiac conditions, including stable or unstable angina pectoris, acute myocardial infarction and cardiogenic shock, the use of this device to treat cardiac conditions other than stable angina pectoris is not covered, since only that use has developed sufficient evidence to demonstrate its medical effectiveness. Non-coverage of hydraulic versions of these types of devices remains in force.

Indications and Limitations of Coverage

B. Nationally Covered Indications

Effective for services performed on or after July 1, 1999, coverage is provided for the use of ECP for patients who have been diagnosed with disabling angina (Class III or Class IV, Canadian Cardiovascular Society Classification or equivalent classification) who, in the opinion of a cardiologist or cardiothoracic surgeon, are not readily amenable to surgical intervention, such as PTCA or cardiac bypass, because:

1. Their condition is inoperable, or at high risk of operative complications or post-operative failure;
2. Their coronary anatomy is not readily amenable to such procedures; or
3. They have co-morbid states which create excessive risk.

A full course of therapy usually consists of 35 one-hour treatments, which may be offered once or twice daily, usually 5 days per week. The patient is placed on a treatment table where their lower trunk and lower extremities are wrapped in a series of three compressive air cuffs which inflate and deflate in synchronization with the patient's cardiac cycle.

During diastole the three sets of air cuffs are inflated sequentially (distal to proximal) compressing the vascular beds within the muscles of the calves, lower thighs and upper thighs. This action results in an increase in diastolic pressure, generation of retrograde arterial blood flow and an increase in venous return. The cuffs are deflated simultaneously just prior to systole, which produces a rapid drop in vascular impedance, a decrease in ventricular workload and an increase in cardiac output.

The augmented diastolic pressure and retrograde aortic flow appear to improve myocardial perfusion, while systolic unloading appears to reduce cardiac workload and oxygen requirements. The increased venous return coupled with enhanced systolic flow appears to increase cardiac output. As a result of this treatment, most patients experience increased time until onset of ischemia, increased exercise tolerance, and a reduction in the number and severity of anginal episodes. Evidence was presented that this effect lasted well beyond the immediate post-treatment phase, with patients symptom-free for several months to two years. This procedure must be done under direct supervision of a physician.

C. Nationally Non-Covered Indications

All other cardiac conditions not otherwise specified as nationally covered for the use of ECP remain nationally non-covered.

(This NCD last reviewed March 2006.)

Claims Processing Instructions

TN 898 (Medicare Claims Processing) [↗](#)

Transmittal Information

Transmittal Number

50

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R50NCD.pdf> [↗](#)

Revision History

01/2024 - Transmittal 12318 issued October 19, 2023, is being rescinded and replaced by Transmittal 12441, dated January 3, 2024, to add FISS to BR 13390.11 and to add clarifying verbiage to business requirement 13390.12. All other information remains the same. ([TN 12441](#) [↗](#)) (CR13390)

10/2023 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 12318](#) [↗](#)) (CR13390)

05/2023 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determination (NCDs). No policy is being changed as a result of these updates. ([TN 12017](#) [↗](#)) (CR13166)

04/2023 - Transmittal 11884 issued March 01, 2023, is being rescinded and replaced by Transmittal 11952, dated, April 12, 2023, to remove the A/B MACs (Part A) and FISS from BR 13070.1 and to revise the NCD 20.4 Implantable Automatic Defibrillators (ICDs) spreadsheet. All other information remains the same. ([TN 11952](#) [↗](#)) (CR13070)

02/2023 - The purpose of the Change Request (CR) is to provide a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes

are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 11832](#)) (CR13070)

08/2022 - Transmittal 11545 dated August 5, 2022, is being rescinded and replaced by Transmittal 11584, dated, August 31, 2022. ([TN 11584](#)) (CR12822)

08/2022 - The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 conversions and other coding updates to specific NCDs. ([TN 11545](#)) (CR12822)

06/2022 - Transmittal 10832, dated June 2, 2021, is being rescinded and replaced by Transmittal 11453, dated, June 10, 2022, to revise NCD 90.2, NGS, revises business requirement 12124.2 and 12124.2.1 and its associated spreadsheet of coding by retaining all ICD-10 NOC diagnosis codes proposed for deletion effective July 1, 2022. ([TN 11453](#)) (CR12124)

06/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10832](#)) (CR12124)

05/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10804](#)) (CR12124)

03/2021 - This Change Request (CR) constitutes a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 10624](#)) (CR12124)

12/2015 - This change request (CR) is the 3rd maintenance update of ICD-10 conversions/updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, & CR9087. Some are the result of revisions required to other NCD-related CRs released separately that included ICD-10 coding. Implementation date: 01/04/2016 Effective date: 10/1/2015. ([TN 1580](#)) (CR9252)

08/2015 - This change request (CR) is the 3rd maintenance update of ICD-10 conversions/updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, & CR9087. Some are the result of revisions required to other NCD-related CRs released separately that included ICD-10 coding.

These updates do not expand, restrict, or alter existing coverage policy. Implementation date: 01/04/2016 Effective date: 10/1/2015. ([TN 1537](#)) (CR 9252)

05/2014 - CMS translated the information for this policy from ICD-9-CM/PCS to ICD-10-CM/PCS according to HIPAA standard medical data code set requirements and updated any necessary and related coding infrastructure. These updates do not expand, restrict, or alter existing coverage policy. Implementation date: 10/06/2014 Effective date: 10/1/2015. ([TN 1388](#)) (CR 8691)

01/2013 - CMS translated the information for this policy from ICD-9-CM/PCS to ICD-10-CM/PCS according to HIPAA standard medical data code set requirements and updated any necessary and related coding infrastructure. These updates do not expand, restrict, or alter existing coverage policy. Implementation date: 04/01/2013 Effective date: 10/1/2015. ([TN 1165](#)) (CR 8109)

03/2006 - Current coverage remains in effect. Effective Date: 03/20/2006 Implementation Date: 04/03/2006. ([TN 50](#)) (CR 4350)

10/2001 - Amended to indicate that policy only pertains to ECP devices intended for treatment of cardiac conditions. Effective and implementation dates 11/15/2001. ([TN 146](#)) (CR 1884)

02/2000 - Changed acronym from EECP to ECP, removed requirement limiting coverage to specific ECP systems, and changed CPT code. Effective and implementation dates 04/01/2000. ([TN 122](#)) (CR 1087)

07/1999 - Changed CPT code. Effective date 07/01/1999. (TN 118)

04/1999 - Revised existing noncoverage policy to limited coverage for use in patients with stable angina pectoris and designated CPT code for billing. Effective date 07/01/1999. (TN 111)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for External Counterpulsation \(ECP\) Therapy \(CAG-00002N\)](#)
- [First reconsideration for External Counterpulsation \(ECP\) Therapy \(CAG-00002R\)](#)

- [Second reconsideration for External Counterpulsation \(ECP\) Therapy \(CAG-00002R2\)](#) 

Additional Information

Other Versions

Title	Version	Effective Between	
External Counterpulsation (ECP) Therapy for Severe Angina	2	03/20/2006 - N/A	You are here
External Counterpulsation (ECP) for Severe Angina	1	11/15/2001 - 03/20/2006	View